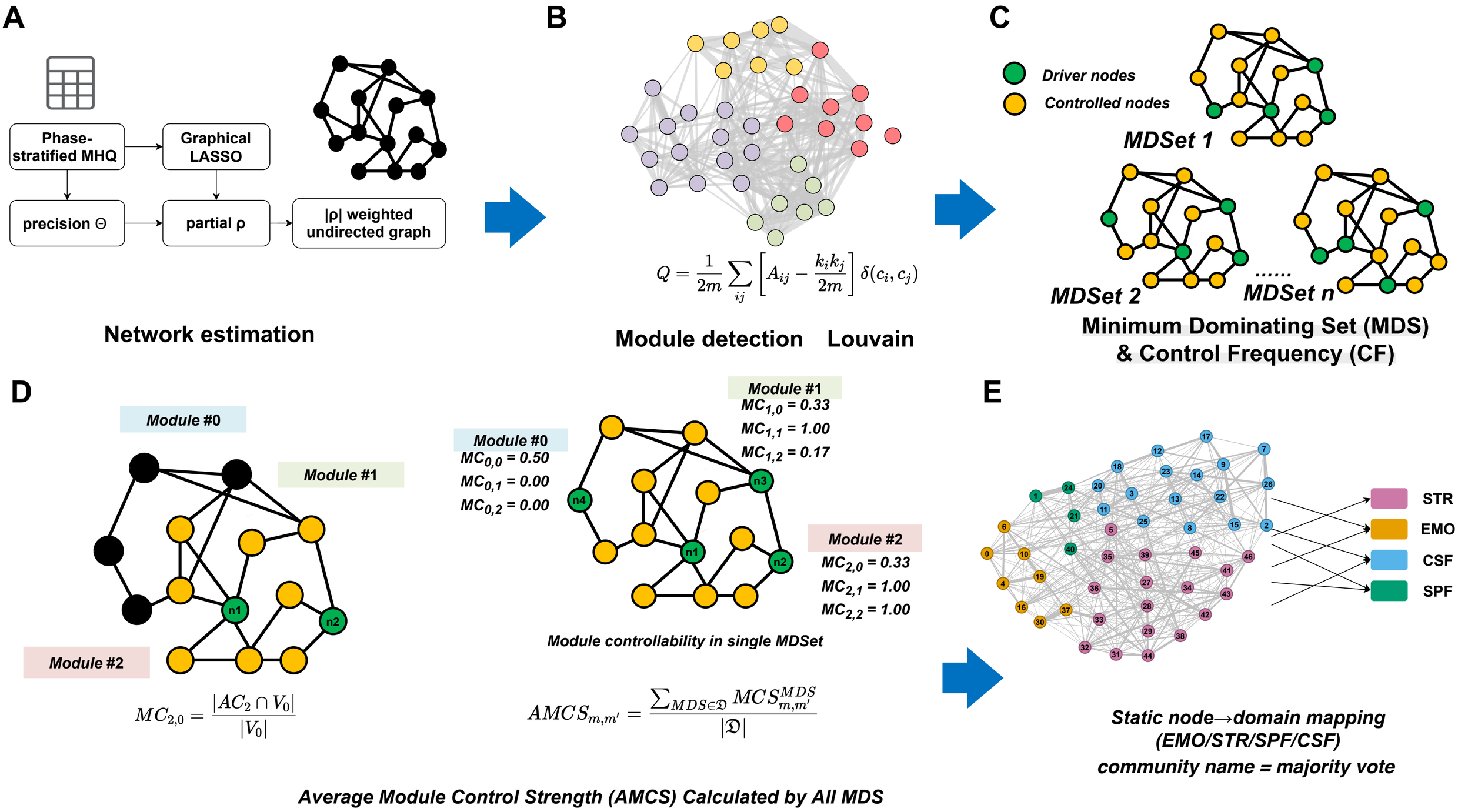
A
Phase-stratified MHQ
Graphical LASSO
precision Θ
partial ρ
|ρ| weighted undirected graph
Network estimation
B
$Q = \frac{1}{2m}\sum_{ij}\left[A_{ij} - \frac{k_i k_j}{2m}\right]\delta(c_i, c_j)$
Module detection Louvain
C
Driver nodes
Controlled nodes
MDSet 1
MDSet 2
.....
MDSet n
Minimum Dominating Set (MDS) & Control Frequency (CF)
D
Module #0
Module #1
Module #2
n1
n2
n3
n4
$MC_{0,0} = 0.50$
$MC_{0,1} = 0.00$
$MC_{0,2} = 0.00$
$MC_{1,0} = 0.33$
$MC_{1,1} = 1.00$
$MC_{1,2} = 0.17$
$MC_{2,0} = 0.33$
$MC_{2,1} = 1.00$
$MC_{2,2} = 1.00$
Module controllability in single MDSet
$MC_{2,0} = \frac{|AC_2 \cap V_0|}{|V_0|}$
$AMCS_{m,m'} = \frac{\sum_{MDS\in\mathfrak{D}} MCS^{MDS}_{m,m'}}{|\mathfrak{D}|}$
Average Module Control Strength (AMCS) Calculated by All MDS
Module-level control
E
STR
EMO
CSF
SPF
Static node→domain mapping (EMO/STR/SPF/CSF)
community name = majority vote
Domain alignment & naming